

3. (2.5 points)

Given the following information as of December 31, 2015:

<u>Accident Year</u>	<u>On-level Premiums</u>	<u>Cumulative Paid Loss</u>	<u>Fitted Paid Emergence Pattern</u>
2012	\$500,000	\$210,000	65%
2013	600,000	150,000	40%
2014	550,000	70,000	20%
2015	650,000	30,000	10%

Cape Cod Method

- Parameter standard deviation: 250,000
- Process variance/mean scale parameter (σ^2): 4,000

LDF Method

- Parameter standard deviation: 325,000
- Process variance/mean scale parameter (σ^2): 4,500

a. (1.25 points)

Calculate the total standard deviation of the total loss reserve indication resulting from the Cape Cod method.

b. (0.75 point)

Calculate the total standard deviation of the total loss reserve indication resulting from the LDF method.

c. (0.5 point)

Explain why σ^2 for the LDF method is higher than σ^2 for the Cape Cod method.

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QUESTION 3															
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as: Chain ladder, Cape Cod, Chain ladder plus calendar-year effects, Bornhuetter-Ferguson.														
SAMPLE ANSWERS															
Part a: 1.25 points															
<i>Sample Answer 1</i> <table> <tr> <td>On lvl prem × emergence</td><td></td></tr> <tr> <td>Used up premium</td><td>CC Reserve = prem × ELR × (1- emergence)</td></tr> <tr> <td>2012 500K × .65 = 325,000</td><td>500,000 × .622 (1 - .65) = 108,850</td></tr> <tr> <td>2013 240,000</td><td>223,920</td></tr> <tr> <td>2014 110,000</td><td>273,650</td></tr> <tr> <td>2015 <u>65,000</u></td><td><u>363,570</u></td></tr> <tr> <td>total 740,000</td><td>Total = 970,320</td></tr> </table> $ELR = \frac{210 + 150 + 70 + 30}{740}$ $ELR = .622$ $process\ std\ dev = \sqrt{4,000 \times 970,320} = 62,300$ $total\ std\ dev = \sqrt{62,300^2 + 250,000^2} = 257,646$ <i>Sample Answer 2</i> CC ELR = $\sum rptd\ Loss / \sum Adj.\ EP \times \% rptd = [460,000 / (500,000(.65) + 600,000(.4) + 550,000(.2) + 650,000(.1))] = 460,000 / 740,000 = .6216$ CC Res = $Adj.\ EP \times ELR \times \% unrptd = 500,000(.6216)(1-.65) + 600,000(.6216)(1-.4) + 550,000(.6216)(1-.2) + 650,000(.6216)(1-.1) = 969,696$ Process Variance = $\sigma^{2R} = 4,000(969,696) = 3,878,784,000$ Total Variance = $Process\ Var + Parameter\ Var = 3,878,784,000 + 250,000^2 = 6.6378 \times 10^{10}$ Total Stdev = $\sqrt{6.6378 \times 10^{10}} = 257,641$		On lvl prem × emergence		Used up premium	CC Reserve = prem × ELR × (1- emergence)	2012 500K × .65 = 325,000	500,000 × .622 (1 - .65) = 108,850	2013 240,000	223,920	2014 110,000	273,650	2015 <u>65,000</u>	<u>363,570</u>	total 740,000	Total = 970,320
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Part b: 0.75 point

Sample Answer 1

$$LDF \text{ reserve} = \frac{\text{paid loss}}{\text{emergence}} - \text{pd loss}$$

$$2012 \ 210,000 / .65 - 210,000 = 113,077$$

$$2013 \quad \quad \quad 225,000$$

$$2014 \quad \quad \quad 280,000$$

$$2015 \quad \quad \quad \underline{270,000}$$

$$\text{total} \quad \quad \quad 888,077$$

$$\text{process std dev} = \sqrt{4,500(888,077)} = 63,217$$

$$\text{total std dev} = \sqrt{62,217^2 + 325,000^2} = 331,091$$

Sample Answer 2

$$LDF \text{ Res} = Ck(CDF-1), \text{ CDF} = 1 / \% \text{ unpaid}$$

$$LDF \text{ Res} = 210,000(1.538-1) + 150,000(2.5-1) + 70,000(5-1) + 30,000(10-1) = 888,077$$

$$\text{Process Variance} = \sigma^{2R} = 4,500(888,077) = 3,996,346,154$$

$$\text{Total Variance} = \text{Process Var} + \text{Parameter Var} = 3,996,346,154 + 325,000^2 = 1.09 \times 10^{11}$$

$$\text{Total Stdev} = \sqrt{1.09 \times 10^{11}} = 331,091$$

Part c: 0.5 point

Sample Answer 1

σ^2 higher for LDF as more parameters in LDF method
& σ^2 calc divides by n-p (# data pts in Δ - # parameters)

Sample Answer 2

The σ^2 for the LDF method is likely higher than for the Cape Cod method because we need to fit fewer parameters for the Cape Cod method (1 parameter for the ELR and 1 each for ω and θ) -> 3 parameters. Whereas the LDF method requires a parameter for every AY (4 + ω and θ) -> 6 parameters. This would lead to overfitting. Can also be seen in the formula for $\sigma^2 = [1 / n-p] \times \sum [(c - \mu)^2 / \mu]$ where c -> actual incremental value, μ -> fitted value. As p increases, the denominator becomes smaller -> σ^2 increases.

Sample Answer 3

$$\sigma^2 = [1 / n-p] \times \sum r^2$$

Since LDF uses more parameters than Cape Cod, it has a higher σ^2 since σ^2 penalizes for using too many parameters (by dividing by (n-p)).

Sample Answer 4

$$\sigma^2 = [1 / n-p] \times \sum [(\text{actual} - \text{expected})^2 / \text{expected}]$$

σ^2 is calculated with the number of parameters (p in the above) and since the LDF method uses more parameters than Cape Cod, the resulting σ^2 is larger.

EXAMINER'S REPORT

Overall, candidates demonstrated a strong understanding of the Cape Cod and LDF methods and how to use those methods to calculate reserves. Additionally, candidates demonstrated a strong understanding of the variance of a reserve estimate; particularly that it consists of two parts –

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process and parameter variance.

Many candidates struggled with part c. Often, candidates responded to part c. as if the question were referring to total variance of the reserve estimate. Instead, the question was referring to the given Cape Cod and LDF σ^2 parameters.

Part a

The majority of candidates achieved full credit on this part or made minimal errors.

Candidates were expected to know:

- How to calculate the Cape Cod ELR (including used-up premium);
- The Cape Cod method for reserves;
- That process variance is the reserve estimate multiplied by σ^2 ; and
- That total variance (and hence total standard deviation) is the sum of both process and parameter variance.

Common errors for part a. were:

- Calculation errors for used-up premium;
- Using ultimates instead of reserves in the calculation of process variance;
- Calculating reserves as $EP \times ELR - \text{paid to date}$;
- Errors resulting from mismatch in scale of figures in formulas (i.e., when converting figures to thousands); and
- Not squaring the parameter standard deviation provided when calculating the formula for total standard deviation.

Part b

The majority of candidates achieved full credit on this part or made minimal errors.

In addition to knowledge of total standard deviation demonstrated in part a., candidates were expected to know the LDF method formula for reserves.

Common errors for part b. were:

- Calculation errors for reserves;
- Using ultimates instead of reserves in the calculation of process variance;
- Errors resulting from mismatch in scale of figures in formulas (i.e., when converting figures to thousands); and
- Not squaring the parameter standard deviation provided when calculating the formula for total standard deviation.

Part c

In general, candidates did not perform well on part c.

To receive full credit, candidates were expected to know:

- That there are more parameters to estimate when using the LDF method compared to the Cape Cod method; and
- That the formula for approximating σ^2 penalizes over-parameterization by including $(n-p)$ in the denominator (i.e., more parameters means a larger scaling factor, all else being equal).

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Common errors for part c. were:

- Not describing the calculation for approximating σ^2 , particularly the penalization for number of parameters;
- Many candidates mentioned that the Cape Cod method uses more information than the LDF method (e.g. premium). This reduces the total variance of the reserve estimate. However, the question refers to the σ^2 process variance/mean scale parameter.